

RESEARCH ARTICLE

Title: Women's Reproductive Health and Agency After the 2006 Bantul Earthquake: A Retrospective Phenomenological Study

Dhesi Ari Astuti^{*1,2}, Setya Putri Rahayu^{2,3}, Nurul Kurniati³ and Elly Naila Fauziah²

¹ Midwifery Study Program, Faculty of Health Sciences, Universitas 'Aisyiyah Yogyakarta, Yogyakarta, Indonesia

² Center for Women, Family, and Disaster Studies, Universitas 'Aisyiyah Yogyakarta, Yogyakarta, Indonesia

³ Universitas 'Aisyiyah Yogyakarta, Yogyakarta, Indonesia

*Address correspondence to Dhesi Ari Astuti at the Midwifery Study Program, Faculty of Health Sciences, Universitas 'Aisyiyah Yogyakarta, Jl. Siliwangi (Ring Road Barat) No. 63, Nogotirto, Gamping, Sleman, Yogyakarta 55292, Indonesia; Tel: +62 858-6887-7555 / +62 877-3904-7482; E-mail: dhesi@unisayogya.ac.id

Background: Women's reproductive health needs and caregiving responsibilities may become harder to meet when disasters disrupt services, shelter, transport, water, sanitation, and household resources.

Objective: To explore how gendered social roles and structural conditions shaped reproductive health vulnerabilities among women who survived the 2006 Bantul earthquake and how women exercised agency during emergency and recovery periods.

Methods: A qualitative retrospective-phenomenological study involved 10 adult women from earthquake-affected areas of Bantul, Yogyakarta, Indonesia. Purposively sampled participants completed semi-structured online interviews between January and March 2026, lasting 45-60 minutes. Verbatim transcripts were examined using reflexive thematic analysis with a hybrid inductive-deductive approach supported by NVivo. Participant feedback on transcripts and preliminary interpretations was used to strengthen credibility.

Results: Three overarching themes were identified: layered reproductive health vulnerabilities; gendered roles and structural barriers; and women's agency and community-based self-reliance. Participants described menstrual hygiene difficulties, disrupted maternal and reproductive health access, reduced privacy, caregiving while injured or postpartum, misinformation, and administrative or logistical barriers. They also described family and community support, livelihood recovery, reproductive decision-making, spatial adaptation, aid advocacy, and social or spiritual coping.

Conclusion: Women's accounts reflected both vulnerability and active agency. Disaster nursing and midwifery planning should recognize reproductive-health continuity, privacy, caregiving burdens, reliable risk communication, and women's participation in local response and recovery while maintaining institutional responsibility for essential services.

Keywords: Disaster Nursing; Earthquakes; Women's Health; Reproductive Health; Midwifery; Qualitative Research; Disasters; Community Resilience

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1. INTRODUCTION

Disasters do not affect populations uniformly. Although an earthquake is a biophysical event, its health and social consequences are shaped by pre-existing inequalities, social roles, economic resources, and access to services. Women may face disproportionate health consequences because reproductive needs and unpaid caregiving responsibilities continue during disruption, while opportunities to seek care, privacy, sanitation, and protection may become constrained [1-6]. Menstrual hygiene, pregnancy, postpartum care, breastfeeding, contraception, and personal safety can therefore become difficult to maintain when health infrastructure and household resources are disrupted [6-10].

The 27 May 2006 Yogyakarta earthquake caused extensive mortality, injury, and infrastructure damage, with Bantul Regency among the most severely affected areas [11,12]. The event disrupted health services, housing, transport, water and sanitation, and local livelihoods. Subsequent work on gender-sensitive disaster risk reduction in Indonesia has emphasized that preparedness and response systems should recognize the different social positions and responsibilities of women and men and should involve women directly in planning and decision-making [13-15].

International humanitarian guidance similarly identifies sexual and reproductive health as an essential component of emergency response. The Minimum Initial Service Package for sexual and reproductive health prioritizes life-saving reproductive health actions at the onset of a crisis, while Indonesian guidance on the minimum reproductive health service package in emergencies emphasizes continuity of maternal, newborn, family-planning, and protection services [16,17]. The Sphere standards and the Sendai Framework-related guidance also stress safe water, sanitation, shelter, health services, community participation, and protection of dignity during emergencies [18,19].

For nursing and midwifery, these disruptions are clinically and operationally important because women may seek help across community, primary-care, emergency, referral, and shelter-linked services. Disaster-nursing literature consistently identifies preparedness, communication, ethical practice, psychosocial readiness, coordination, and continuity of care as core capabilities during emergencies [20-24]. In reproductive health, evidence from humanitarian settings also shows that service use and intervention effectiveness depend not only on the availability of clinical services but also on access, referral pathways, communication, and the organization of care during crisis conditions [25-28]. These considerations make survivors' accounts relevant to nurses and midwives who participate in emergency assessment, maternal and reproductive health care, community outreach, risk communication, and recovery planning.

However, women's experiences in disasters cannot be understood only through a vulnerability lens. Research on gender and disasters increasingly shows that women also act as planners, caregivers, economic actors, organizers, and local risk managers. Their responses may include mobilizing social capital, reorganizing domestic space, sustaining food

systems, negotiating services, and creating informal networks of psychosocial support [4,5,11]. Recognizing this agency is important, but it should not be used to transfer responsibility for essential services from institutions to affected communities.

Most published disaster-health literature describes immediate needs, service disruptions, or population-level outcomes. Less attention has been given to how women retrospectively connect reproductive health needs with domestic responsibilities, institutional barriers, and practical forms of agency within a specific Indonesian earthquake context. A survivor-centered qualitative account can therefore complement epidemiological and service data by showing how needs were experienced, negotiated, and remembered at household and community levels. Such evidence is not intended to establish prevalence or causal effects; rather, it can identify issues that disaster-nursing, midwifery, and community-health planning may need to anticipate.

Nearly two decades after the 2006 earthquake, survivors' retrospective accounts offer an opportunity to examine how reproductive health needs, gendered responsibilities, institutional barriers, and self-directed recovery strategies were remembered and interpreted over time. This study therefore explored how gendered social roles and structural conditions shaped layered reproductive health vulnerabilities among women survivors of the Bantul earthquake and how women exercised agency to manage health, caregiving, household, and community needs during emergency and recovery periods.

2. MATERIALS AND METHODS

2.1. Study Design and Setting

This study used a qualitative retrospective-phenomenological design to explore how women survivors interpreted and reconstructed their experiences of reproductive health, caregiving, and recovery after the 2006 Bantul earthquake. The study was conducted with residents of earthquake-affected areas in Bantul Regency, Special Region of Yogyakarta, Indonesia. The design was selected to foreground the meaning participants assigned to events that occurred during the emergency and recovery periods while acknowledging that these accounts were reconstructed many years after the disaster.

2.2. Participants and Sampling

Participants were recruited purposively. Eligibility criteria were adult women who had experienced the 2006 Bantul earthquake and were living in affected areas of Bantul at the time of recruitment. Ten women participated. Their ages at the time of the disaster ranged from 29 to 44 years, and their reproductive situations included pregnancy, breastfeeding, menstruation, perimenopause, contraceptive use, and caregiving for children of different ages. Participant characteristics are summarized in Table 1. The study sought detailed, information-rich accounts across varied reproductive and caregiving circumstances rather than statistical representativeness. No quantitative sample-size calculation was used and no formal saturation threshold is

claimed. The focused study aim, specificity of the participant group, and depth of the 45-60-minute interviews are therefore considered when interpreting sample adequacy, **Table 1.** Participant characteristics during the 2006 earthquake.

consistent with qualitative-method guidance on information power and saturation [29,30].

Participant	Age (years)	Occupation / socioeconomic role	Living children	Reproductive phase / dominant caregiving role	Contraceptive status
I1	44	Teacher / educator	2	Perimenopause; care of adolescents	Not applicable
I2	40	Housewife	1	Care of toddler	Not used
I3	41	Market vendor	2	Care of school-age children and adolescents	IUD
I4	29	Farmer / farm worker	1	Third-trimester pregnancy; high-risk pregnancy	Not applicable
I5	33	Housewife	3	Menstruation; breastfeeding; care of toddlers and school-age children	Not used
I6	43	Housewife	2	Care of school-age children	3-month injectable contraceptive
I7	38	Farmer	3	Breastfeeding; care of infant, toddler, and school-age children	3-month injectable contraceptive
I8	38	Housewife	2	Breastfeeding; care of toddlers and school-age children	Not used
I9	33	Merchant	2	Care of toddlers and school-age children	Oral contraceptive
I10	43	Teacher / educator	2	Care of school-age children and adolescents	Not used; husband had vasectomy

Source: Primary data. IUD: intrauterine device.

2.3. Data Collection

Semi-structured in-depth interviews were conducted from January to March 2026 using Zoom Meetings and lasted approximately 45-60 minutes. The interview guide was developed from disaster-sociology, gender-vulnerability, and community-health literature and was reviewed by colleagues with expertise in gender studies and community health for relevance, trauma sensitivity, and ethical appropriateness. Questions explored three domains: reproductive health vulnerabilities; gender roles and structural barriers; and women's agency and self-directed recovery strategies.

Some participants in rural areas required assistance with digital access. Local undergraduate students served as field assistants to set up devices, support connection stability, and reduce technology-related difficulties. After a secure connection was established, assistants left the immediate interview space to preserve visual and auditory privacy. Interviews were led by women health professionals and academics familiar with the local Javanese context. This positionality facilitated rapport but also created the possibility that shared cultural assumptions could influence interpretation; reflexive discussion was therefore incorporated during analysis.

2.4. Data Analysis and Trustworthiness

Interviews were audio-recorded with participant permission and transcribed verbatim. Data were analyzed using Braun and Clarke's thematic analysis approach [31,32]. Analysis combined inductive attention to participants' own language with deductive attention to the study's three domains. Researchers repeatedly read transcripts, generated initial codes, grouped related codes, reviewed candidate themes against the dataset, and refined theme definitions. Final

coding and theme consolidation were performed by Dhesi Ari Astuti. NVivo software was used to organize coded material, retrieve excerpts, and support visual mapping; it was not treated as an independent measure of theme importance. Because the analysis was reflexive, numerical inter-rater agreement and code frequencies were not treated as indicators of validity; the emphasis remained on interpretive engagement with the dataset and the coherence of themes across participants' accounts [33].

To strengthen credibility, transcripts and summaries of preliminary interpretations were discussed again with participants. Field assistants supported follow-up contact and, when needed, read key material aloud during face-to-face visits. The research team used these responses to check whether accounts had been represented recognizably while retaining responsibility for the final analytic interpretation. Reporting was reviewed against the principles of the Consolidated Criteria for Reporting Qualitative Research (COREQ) [34]. Trustworthiness was further supported by making the analytic steps explicit, revisiting candidate themes against the full dataset, incorporating reflexive discussion, and distinguishing participant accounts from claims that would require external verification. These practices are consistent with broader recommendations for transparency and credibility in qualitative thematic analysis [35,36].

2.5. Ethical Considerations

Ethical approval was obtained from the Health Research Ethics Committee of Universitas 'Aisyiyah Yogyakarta (No. 4965/KEP-UNISA/XII/2025) for the protocol titled 'Qualitative Study: Memories of the 2006 Earthquake among Adult Women in Bantul'. Data collection occurred from January to March 2026, within the approval period. Verbal

informed consent was obtained in two stages. The first consent was obtained during an in-person explanation of the study, and consent was reconfirmed immediately before the online interview. Participants were informed about the study purpose and procedures, confidentiality, voluntary participation, and their right to refuse or withdraw without consequences. Interviews were conducted in a manner intended to minimize distress when sensitive memories were discussed. Digital files were stored in encrypted storage accessible only to the core research team. The research was funded by the Center for Women, Family, and Disaster Studies (PSPKB), Universitas 'Aisyiyah Yogyakarta.

Table 2. Summary of overarching themes and key findings.

Overarching theme	Dimension	Key categories / findings
Layered reproductive health vulnerabilities	Socioeconomic	Menstrual hygiene and clean-water difficulties; loss of infant supplies; food and nutrition insecurity; self-neglect while prioritizing family needs.
Layered reproductive health vulnerabilities	Social roles	Caregiving while injured or postpartum; breastfeeding-related constraints during evacuation; persistent trauma-related symptoms; reduced privacy for postpartum care and family life.
Layered reproductive health vulnerabilities	Security / health services	Blocked evacuation routes; overloaded health facilities; delayed treatment; misinformation; disrupted or uncertain access to contraception and preventive services.
Gendered roles and structural barriers	Socioeconomic	Domestic exposure during early-morning household work; exclusion or delay in housing assistance for some young households; reconstruction shortfalls and informal borrowing; disrupted market access.
Gendered roles and structural barriers	Social roles	Expectation of maternal self-sacrifice; continued unpaid care work during injury or postpartum recovery; intergenerational caregiving responsibilities; dual public and domestic roles.
Gendered roles and structural barriers	Security / administration	Reported irregularities in aid distribution; congestion and delays in formal services; limited privacy in communal shelters; dependence on local intermediaries when formal response was delayed.
Women's agency and community-based self-reliance	Socioeconomic	Restarting trade and household food production; sharing farm produce; improvising cooking facilities; mobilizing household and family resources; maintaining reproductive decision-making where possible.
Women's agency and community-based self-reliance	Social roles	Use of extended-family networks; informal childcare arrangements; collective humor and social gatherings; adaptive relocation; modification of sleeping and household arrangements for safer evacuation.
Women's agency and community-based self-reliance	Security / community leadership	Participation in local aid negotiations; organization of sanitation and logistics; women's community and religious groups as psychosocial support; household redesign with multiple evacuation exits.

Source: Primary data; categories condensed from the full thematic coding matrix.

3.2. Theme 1: Layered Reproductive Health Vulnerabilities

Participants described reproductive health vulnerability as a combination of bodily needs, disrupted services, constrained living conditions, and caregiving demands. Menstruation in evacuation settings was particularly difficult when sanitary pads, clean water, dry clothing, and private washing facilities were unavailable. One participant recalled having no menstrual supplies during the early emergency period:

"At that time, I was still menstruating... I didn't have any sanitary pads; I didn't know what to do—there was no aid... until I was completely soaked..." (I5)

Pregnancy, postpartum recovery, and breastfeeding created additional demands. Participants described loss of infant clothing and supplies, exposure to dust and rain, limited privacy in communal tents, and difficulty protecting infants while they themselves were injured or recovering. A participant who gave birth around the disaster period

3. RESULTS

3.1. Overview of Themes

The analysis generated three overarching themes: [1] layered reproductive health vulnerabilities; [2] gendered roles and structural barriers; and [3] women's agency and community-based self-reliance. These themes were interrelated rather than sequential. Participants often described the same event as both a source of vulnerability and a context in which they developed practical responses. Table 2 summarizes the principal findings across socioeconomic, social-role, and security/administrative dimensions.

described an overcrowded clinical environment and a distressing experience of labor during the emergency:

"I was screaming at the nurse... trying to ask for help, the baby was about to come out, but no one was coming near me... finally, I screamed that the baby had already come out... right on the bed..." (I4)

Participants also described delayed treatment because roads were blocked, vehicles were limited, or facilities were crowded. Some relied on home or traditional remedies when formal services were difficult to reach. Contraceptive access varied. A few participants retained access through community health services or by keeping oral contraceptives at home, whereas others described uncertainty about availability during the emergency period.

Psychological distress was intertwined with these physical and reproductive concerns. Participants reported persistent fear of buildings, emotional numbness, anxiety about

children, and long-lasting memories of injury and loss. These narratives are reported as trauma-related experiences rather than clinical diagnoses because no standardized diagnostic assessment was conducted.

3.3. Theme 2: Gendered Roles and Structural Barriers

Women's exposure to risk was shaped by the timing and location of domestic work. Several participants were preparing breakfast or caring for children inside the home when the earthquake struck, while some male relatives were already outdoors. One participant recalled being in the kitchen immediately before a structural collapse:

"It's time to make breakfast... even a moment's delay and that concrete pillar would have already collapsed."
(110)

Maternal responsibility was frequently described as an immediate obligation to return for or physically protect children even when doing so increased personal risk. One participant described going back into a damaged house for her child despite being injured:

"The children are still inside... their father went out on his own, I went back in... the roof collapsed, and my head kept bleeding..." (13)

Structural barriers continued during recovery. Participants described difficulties related to household registration, housing-assistance eligibility, limited reconstruction funds, delayed health support, and geographic inequality in aid distribution. These accounts represent participants' perceptions and experiences; the study did not independently audit historical administrative records or establish misconduct by specific institutions. Accordingly, the findings are interpreted as reported administrative and access barriers rather than verified corruption.

Communal shelter also created privacy constraints for postpartum care, breastfeeding, rest, and married couples' intimate life. Participants emphasized that crowded tents required constant adaptation. In addition, women often continued cooking, caring for children and older relatives, and providing emotional support while managing their own injuries or stress.

3.4. Theme 3: Women's Agency and Community-Based Self-Reliance

Participants did not describe themselves only as recipients of assistance. They also developed practical strategies to maintain health and household functioning. Women restarted food preparation and market activity, shared produce, improvised stoves from available materials, and used extended-family networks to reduce pressure on crowded shelters. These actions supported nutrition, household liquidity, and continuity of caregiving during periods when formal systems were disrupted.

"The rice fields still have the crops that were there before. There's water spinach, there's chili... well, just take them for the community..." (13)

Agency also appeared in decisions about bodily and reproductive autonomy. Some women maintained

contraception through community health services or by storing oral contraceptives, while another family relied on the husband's prior vasectomy. These strategies should not be interpreted as a substitute for reliable reproductive health services; rather, they illustrate how households attempted to preserve reproductive decision-making under uncertain conditions.

Participants described changes to domestic space that emerged from earthquake experience. Some slept near exits, avoided locked doors, or redesigned homes with more than one exit from a room. One participant described a house reconstructed with multiple doors:

"Her house has so many doors. Each room has two. One in the front, one in the back..." (18)

Women also used social and religious networks as forms of collective coping. Community gatherings, Family Welfare Empowerment (PKK) activities, Yasinan, Asmaul Husna recitation, humor, and mutual assistance were described as ways of reducing isolation and rebuilding social connection. Participants additionally described involvement in neighborhood discussions about aid allocation, sanitation, and local recovery, indicating that women's agency extended beyond the household to community-level organization.

4. DISCUSSION

This study shows that women's experiences after the 2006 Bantul earthquake cannot be adequately characterized as either vulnerability or resilience alone. Reproductive health needs, caregiving responsibilities, shelter conditions, service disruption, and administrative barriers created layered risks, while women simultaneously mobilized household, family, economic, spatial, social, and spiritual resources. The findings therefore support a relational understanding of disaster vulnerability in which agency emerges within, and is constrained by, social and institutional conditions [4,5,11,37].

The first major finding concerns the continuity of reproductive health needs during emergency conditions. Menstruation, pregnancy, postpartum recovery, breastfeeding, contraception, and privacy did not stop when housing and health systems were disrupted. Participants' accounts of menstrual hygiene difficulties, crowded shelters, delayed treatment, and uncertain contraceptive access are consistent with humanitarian evidence showing that sexual and reproductive health services are frequently disrupted during crises and that women may face increased barriers to maternal care and family planning [7,8,16-18,38,39]. These findings reinforce the importance of implementing reproductive health emergency packages from the earliest phase of response rather than treating them as secondary to food, shelter, and trauma care. Evidence from displaced and crisis-affected populations similarly shows that menstrual hygiene depends on the combined availability of menstrual materials, safe water and sanitation, privacy, disposal options, and practical information [40-43]. Pregnancy and the perinatal period may also be affected through service interruption, environmental disruption, stress, and constrained access to routine or urgent care; systematic and

post-disaster studies support treating maternal and reproductive health as an integral component of disaster response rather than a later-stage addition [44-49]. The present study does not test whether any particular intervention would have prevented the difficulties described by participants, but it adds contextual evidence about how multiple needs converged within the same households.

The second major finding concerns the gendered distribution of domestic exposure and unpaid care work. Participants were often inside homes because they were cooking, breastfeeding, or caring for children when the earthquake occurred. The accounts of returning for children, providing care while injured, and continuing community cooking or caregiving during postpartum recovery illustrate how gendered expectations can shape exposure and recovery burden. Similar work has shown that disasters amplify existing gendered divisions of labor and may increase women's physical and psychosocial burdens [3-6,9,37,50,51]. Traditional Javanese housing and kitchen arrangements may also shape spatial exposure, although the present qualitative study cannot quantify whether specific house types increased women's injury risk [52].

Administrative and logistical barriers were also prominent. Participants described delayed or uneven aid, eligibility problems related to family registration, overloaded services, blocked roads, and dependence on local intermediaries. The Java Reconstruction Fund and post-disaster assessments documented the complexity of housing reconstruction and recovery after the 2006 earthquake [12,53]. Our findings add a survivor-centered perspective by showing how such procedures were experienced at household level. However, retrospective interviews cannot establish whether all reported irregularities reflected formal policy, implementation gaps, local discretion, or individual misunderstanding. This distinction is important when interpreting participants' accounts of aid distribution.

A third contribution is the emphasis on women's agency. Participants described reopening trade, sharing agricultural products, organizing food preparation, mobilizing extended-family support, adapting shelter arrangements, participating in neighborhood decisions, and using religious and women's community groups for psychosocial support. These forms of local action resemble findings that women frequently contribute to community resilience through informal care, social capital, and economic activity [5,11,54-57]. Their role in local recovery should therefore be recognized in disaster-health planning, including formal representation in village disaster teams, health clusters, shelter planning, and risk-communication systems.

At the same time, women's agency should not be romanticized. The ability to improvise food systems, self-manage contraception, borrow money, or redesign shelter may reflect resourcefulness, but it can also signal gaps in public support. Agency-based health promotion is most useful when institutions create conditions in which women can participate in decisions and access reliable services, not when communities are expected to compensate for missing infrastructure. This distinction is especially important for maternal and reproductive health, where delayed clinical

care, unsafe sanitation, and interrupted contraception can have serious consequences [16-18,38,39].

The accounts of persistent fear, emotional numbness, sleep-related vigilance, and maternal anxiety also show the close connection between reproductive, caregiving, and mental health. Disaster exposure is associated with substantial psychological morbidity, and pregnancy-related stress can affect maternal well-being [1,2,58]. Because this study did not use diagnostic instruments, these experiences should be understood as trauma-related symptoms rather than confirmed post-traumatic stress disorder. Community and spiritual practices described by participants may have supported coping and social connection, but they should complement rather than replace access to professional mental health care when needed. Post-traumatic growth and positive reappraisal have been documented after severe adversity, although such processes vary considerably between individuals [59]. Large reviews of disaster-exposed populations show wide variation in psychological outcomes and caution against assuming that all survivors develop psychiatric disorders [60,61]. Principles proposed for early and mid-term post-disaster support emphasize safety, calming, efficacy, connectedness, and hope, while contemporary resilience scholarship also treats adaptation as dynamic and context dependent rather than a fixed personal trait [62,63]. These perspectives are compatible with the present decision to report participants' experiences as trauma-related symptoms and coping responses rather than diagnoses.

4.1. Implications for Nursing, Midwifery, and Disaster Health Practice

The findings suggest several practice considerations for nurses, midwives, and interdisciplinary disaster-health teams. First, rapid needs assessment should explicitly include menstruation, pregnancy, postpartum recovery, breastfeeding, contraception, infant care, privacy, and the practical burden of caring for dependents. This does not require treating every reproductive need as a specialist emergency; it requires ensuring that these needs are visible in triage, referral, shelter assessment, community outreach, and supply planning. When clinical services are overloaded or transport is disrupted, clear referral pathways and reliable communication become especially important.

Second, the study supports integrating reproductive-health continuity with core disaster-nursing functions rather than handling it as a separate concern. Reviews of disaster preparedness consistently identify gaps in nurses' disaster knowledge and skills, while more recent syntheses emphasize training, communication, planning, psychological preparedness, and coordinated roles across the disaster cycle [64-69]. In practical terms, preparedness exercises can include scenarios involving postpartum women, breastfeeding mothers, contraception continuity, menstrual hygiene, caregiving responsibilities, and women who cannot easily leave dependents to seek care. The present findings cannot demonstrate that such training improves outcomes, but they identify concrete situations that training and protocols should be able to address.

Third, privacy and dignity should be treated as clinical and safety issues. Participants' accounts linked crowded shelter conditions with difficulties in menstruation, breastfeeding, postpartum care, rest, and intimate family life. Nursing and midwifery input into shelter-health planning can help identify requirements for private assessment, sex-separated sanitation, safe areas for maternal and newborn care, confidential counselling, and communication that does not expose sensitive reproductive information. These measures are particularly relevant when women continue unpaid care work while injured, postpartum, or psychologically distressed.

Fourth, community networks should be viewed as resources for communication and recovery without transferring institutional responsibilities onto women. Participants described extended-family support, food sharing, local gatherings, religious networks, and participation in neighborhood decisions. Disaster research identifies social capital as an important mechanism through which information, mutual aid, collective action, and recovery resources can circulate [70-75]. Nurses and midwives working through primary-care and community systems can potentially use trusted local networks to extend risk communication, identify unmet needs, and support referral. However, informal networks should complement reliable public services; they should not become a rationale for expecting affected households to replace missing health, sanitation, or protection infrastructure.

Finally, women should be included in preparedness and recovery decisions because they hold practical knowledge about household routines, caregiving, privacy, evacuation, and access barriers. The study provides qualitative support for involving women in shelter design, local logistics, aid registration, risk communication, and community recovery discussions. These are implications derived from participants' experiences rather than tested effects, and their implementation should be adapted to local systems, available resources, and the responsibilities of disaster-management and health authorities.

From a disaster-cycle perspective, these findings can be translated into phase-specific questions for practice without assuming intervention effectiveness. During preparedness, nursing and midwifery teams can review whether local plans identify reproductive-health supplies, maternal and newborn referral options, privacy requirements, evacuation barriers for caregivers, and trusted communication channels. During the acute response, brief assessment can make pregnancy or postpartum status, breastfeeding needs, menstrual hygiene constraints, contraception interruption, injuries, dependent-care responsibilities, and distress visible so that urgent clinical or protection needs can be escalated through appropriate pathways. During early recovery, linkage with primary care, reproductive-health services, psychosocial support, and household recovery systems may help identify needs that were less visible during the immediate emergency. The same accounts also caution against equating self-reliance with low need: a woman who improvises food, childcare, transport, or contraception may still be compensating for disrupted services. Accordingly, disaster-

health teams should distinguish between capacity to act and adequacy of formal support. These phase-specific applications are practice propositions grounded in survivors' accounts and related literature, not tested care pathways, and should be evaluated prospectively before being interpreted as evidence of effectiveness.

For nursing and midwifery education, the cases described by participants offer concrete situations for discussion of person-centred assessment, communication, referral, privacy, continuity of reproductive care, psychosocial support, and interprofessional coordination. Simulation or case-based learning can use such situations to examine how clinical priorities intersect with caregiving, shelter conditions, transport, misinformation, and administrative access. Training should avoid presenting women as a uniform risk group; reproductive phase, household responsibilities, available support, and access constraints differed across participants. This approach preserves the study's central finding that vulnerability and agency coexisted and that professional support should respond to both rather than reducing women either to passive victims or to substitutes for functioning services.

4.2. Strengths and Limitations

A strength of this study is its survivor-centered focus on the intersection of reproductive health, gendered responsibilities, institutional access, and self-directed recovery. The long interval since the earthquake also allowed participants to reflect on which adaptations persisted into later life. The study used reflexive thematic analysis, participant feedback, and explicit researcher positionality to support transparency.

Several limitations require caution. First, interviews were conducted almost two decades after the earthquake, creating substantial potential for recall bias, memory reconstruction, and influence from later experiences. Member feedback may improve representational credibility but cannot eliminate this limitation. Second, the sample was small, purposive, and drawn from one district; findings are not intended to estimate prevalence or be statistically generalized. Third, the study included only women survivors and did not triangulate accounts with men, health workers, government officials, or contemporaneous administrative records. Fourth, online interviewing, even with field assistance, may have influenced disclosure or privacy. Fifth, reported symptoms such as fear, respiratory illness, reproductive tract infection risk, or delayed labor were not clinically verified, and causal relationships cannot be inferred from retrospective qualitative narratives. Finally, coding density or software visualization was not interpreted as a quantitative measure of the relative importance of themes. The findings should therefore be read as context-specific retrospective accounts rather than prevalence estimates or causal evidence.

5. CONCLUSION

Women survivors of the 2006 Bantul earthquake described reproductive health vulnerability as an interaction between bodily needs, gendered caregiving responsibilities, disrupted services, shelter conditions, and administrative or logistical barriers. They also described agency through household

adaptation, community organization, economic recovery, reproductive decision-making, spatial preparedness, and social or spiritual support. For disaster nursing and midwifery, the findings point to the value of making menstrual, maternal, postpartum, breastfeeding, contraception, privacy, caregiving, and psychosocial needs visible in early assessment and continuity-of-care planning. Women's participation in shelter, logistics, risk communication, and recovery planning may improve the contextual relevance of response arrangements, but this qualitative study does not test the effectiveness of specific interventions. Recognizing women's agency should strengthen, rather than replace, institutional responsibility for equitable health services and protection.

AUTHORS' CONTRIBUTIONS

D.A.A.: Conceptualization, Methodology, Investigation, Formal Analysis, Interpretation, Writing - Original Draft, Writing - Review & Editing, Supervision, and Project Administration. S.P.R.: Methodology, Investigation, Data Collection, Formal Analysis, Interpretation, and Writing - Review & Editing. N.K.: Investigation, Data Collection, Formal Analysis, Interpretation, and Writing - Review & Editing. E.N.F.: Formal Analysis, Interpretation, Writing - Original Draft, and Writing - Review & Editing. All authors read and approved the final version of the manuscript.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

The study was approved by the Health Research Ethics Committee of Universitas 'Aisyiyah Yogyakarta (No. 4965/KEP-UNISA/XII/2025). Verbal informed consent was obtained from all participants in two stages: during the initial in-person study explanation and again immediately before the online interview. Participation was voluntary, and participants were informed of confidentiality protections and their right to refuse or withdraw without consequences.

HUMAN AND ANIMAL RIGHTS

This was a non-interventional qualitative study involving human participants. The protocol received formal institutional ethics approval as stated above. The ethics approval documented conformity with seven WHO 2011 ethical standards, including informed consent, confidentiality and privacy, risks, scientific value, and equitable assessment, with reference to CIOMS 2016. No animals were involved in the study.

CONSENT FOR PUBLICATION

No directly identifying personal material is included in this manuscript. Participant quotations are presented anonymously using study identifiers. Verbal informed consent for study participation and interview recording was obtained as described above; no photographs, videos, names, or other directly identifying materials are presented.

STANDARDS OF REPORTING

The manuscript was prepared with reference to the Consolidated Criteria for Reporting Qualitative Research

(COREQ) [34]. Sex- and gender-related terminology, analysis, and interpretation were also reviewed in line with SAGER principles. Reporting items not documented in the study records were not inferred or retrospectively invented.

AVAILABILITY OF DATA AND MATERIALS

Anonymized interview transcripts supporting the findings may be available from the corresponding author upon reasonable request, subject to participant confidentiality, ethical commitments, and institutional requirements.

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CONFLICT OF INTEREST

The authors confirm that there is no conflict of interest related to this manuscript.

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